Pranav Venkateswaran

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EDUCATION

Thomas Jefferson High School for Science and Technology, Alexandria, VA **Expected June 2028**
Weighted GPA: 4.41 | SAT: 1520 | Coursework: Web Application Development (in progress), AP CS A, AP Statistics

SOFTWARE PROJECTS

Emercity | Offline-first household emergency app | Code, Architecture notes **2026**
TypeScript, React Native, Expo, SQLite (SQLCipher), Firebase Auth and Firestore, Jest

- Built an iOS/Android app that ranks who in a household needs help first across 6 hazards with a deterministic rule engine: every ranking is explainable, test-reproducible, and conservative when inputs are missing
- Made encrypted SQLite the source of truth: each shared-household write enqueues a sync record in the same database transaction, which drains to Firestore on reconnect, so offline edits are never lost
- Verified with 462 unit tests across 35 suites, including the priority engine against 46 fixture households, plus an emulator-run test suite for the role-based Firestore security rules

Finance Digest | Civic-finance iOS app | Code **2026**
TypeScript, React Native, Expo Router, Jest, Treasury Fiscal Data API, USAspending.gov API

- Built an app that turns income, filing status, and ZIP code into a personal federal tax receipt, state and local tax estimates for all 50 states plus DC, and a map of nearby federal awards
- Routed the tax receipt and what-if budget simulator through one pure allocation engine; parameterized Jest tests assert that line items sum exactly to the tax and every transfer conserves the total
- Validated each Treasury data refresh against the published total; LLM explainers never supply a receipt figure

Biggie's Adventure | Technology Student Association (TSA) Video Game Design team entry | Play **2025**

- Co-developed a 2-player, multi-level C#/Unity platformer on a shared Git codebase and shipped a WebGL build

RESEARCH

Vision-Language Model Evaluation | Independent research, sole author **2026**
"Shape, Seam, or Sense? Probing the Cues Vision-Language Models Use to Reassemble Natural and Biomedical Images"
Under review at an AAAI-27 workshop; paper and code will be released after review

- Built a Python benchmark that cuts images into 6×6 interlocking puzzles under 9 cue conditions in 4 image domains, with a shape oracle (IoU $\geq$ 0.6) guaranteeing one valid answer and a classical seam solver as a non-AI baseline
- Evaluated 4 open 4-12B models locally plus Gemini and GPT-5.5 using per-option probabilities for calibrated scoring; enforced API spend caps in code against a global ledger and re-derived every result with an independent audit script
- Pre-registered the analysis; 2 validity controls changed the conclusions: a color-histogram baseline often solved random-distractor items, and the open model that passed screening fell to chance on content-matched distractors

NASA GeneLab for High Schools (GL4HS) | Capstone Team Lead, NASA Ames (virtual) **June - August 2026**

- Selected for the Capstone Program (capped at 45 teams nationally) and led a team analyzing NASA spaceflight RNA-seq data (OSD-437) with differential expression and DAVID/KEGG pathway enrichment; presented at the GL4HS Showcase
- Poster accepted to the 2026 American Society for Gravitational and Space Research meeting (co-author, co-presenter)

D-Select: ML Peptide Design for TB Breath Diagnostics | Independent student research, co-author **Summer 2026**

- Co-built a pipeline that scores peptides with a PLAPT and DeepPurpose affinity ensemble weighted by breath-compound concentration; the lead peptide reached ROC-AUC 0.955 on held-out synthetic data

George Mason University, Center for Infectious Disease Research | Research Intern **June - August 2025**

- Designed antimicrobial peptides with CAMPR4, AlphaFold, and AutoDock Vina; presented August 2025 (MARS archive)

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, C#, HTML/CSS
Backend & web: Node.js, Express, REST APIs, OAuth 2.0, session auth, role-based access control, SQL transactions, React
Mobile & testing: React Native, Expo, SQLite (SQLCipher), Firebase Auth and Firestore rules, Zustand, Jest
ML & tools: PyTorch, MLX, NumPy, pandas, SciPy, Matplotlib, Git and GitHub, Unity

HONORS & ACTIVITIES

Honors: TSA State Medalist and National Qualifier (2025); Science Olympiad State Finalist and Regional Medalist (2024)
Activities: Research Coordinator, debate team; Loudoun County Youth Advisory Council; EMT-Basic training (2026)